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Course Code: IM244AI

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IV Semester B. E. Sept / Oct 2024 Examinations
DEPARTMENT OF INDUSTRIAL ENGINEERING AND MANAGEMENT
Course: OPERATIONS RESEARCH
(2022 Scheme)

Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.
2. Answer FIVE full questions from Part B. In Part B question number 2 is compulsory. Answer any one full question from 3 and 4, 5 and 6, 7 and 8 and 9 and 10.

PART – A (Objective type for one or two marks)
(True & false and match the following questions are not permitted)

1			Marks	BT	CO
	1.1	List the application of Operations Research in areas of finance.	02	2	1
	1.2	The corner points of the feasible solutions are (0, 0), (3,0), (2,1), (0,7/3) the maximum value of Z = 4x1 + 5x2 is _____.	02	2	1
	1.3	Optimality is indicated for a maximization problem when all elements in the cj – zj rows are _____, while for a minimization problem all elements must be _____.	02	2	1
	1.4	The simplex method examines the _____ points in a systematic manner, repeating the same set of steps of the algorithm until an _____ solution is reached.	02	1	1
	1.5	What is an unbalanced transportation problem? How to resolve it.	02	2	2
	1.6	What is the indication of an alternate solution in an assignment problem?	02	2	2
	1.7	Activities A, B, and C are the immediate predecessors for Y activity. If the earliest finishing times for the three activities are 12, 15, and 10, then what will be the earliest starting time for Y?	02	2	2
	1.8	How does PERT and CPM helps in solving complex business problems?	02	2	2
	1.9	The graphical method is useful for game where payoff matrix is of size _____ or _____.	02	1	1
	1.10	The rules of _____ are used to reduce the size of the _____ matrix.	02	2	2

PART – B (Maximum subdivision is limited to 4 in each question)

		UNIT – I			
2	a	List and explain the different phases of operations research.	08	2	2
	b	A company has two grades of inspectors, I and II to undertake quality control inspection. At least 1, 500 pieces must be inspected in an 8 – hour day. Grade I inspector can check 20 pieces in an hour with an accuracy of 96%. Grade II inspector checks 14 pieces an hour with an accuracy of 92%. Wages of grade I inspector are Rs. 5 per hour while those of grade II inspector are Rs. 4 per hour. Any error made by an inspector costs Rs. 3 to the company. If there are, in all, 10	08	3	3

		grade I inspectors and 15 grade II inspectors in the company, formulate the problem as linear programming problem for the optimal assignment of inspectors that minimize the daily inspection cost.			

		UNIT - II			
3	a	What type of actions to be taken for the following constraints in a standard form of simplex method? i. Less than or equal to (≤) ii. Greater than or equal to (≥)	04	3	3
	b	Solve the following linear programming problem by simplex method: Maximize $Z = 16x_1 + 17x_2 + 10x_3$ subject to the constraints $x_1 + x_2 + 4x_3 \leq 2{,}000$ $2x_1 + x_2 + x_3 \leq 3{,}600$ $x_1 + 2x_2 + 2x_3 \leq 2{,}400$ $x_1 \leq 30$ and $x_1, x_2, x_3 \geq 0$	12	2	3
		OR			
4		Solve the following linear programming problem by dual simplex method: Minimize $Z = 2x_1 + 2x_2 + 4x_3$ Subject to $2x_1 + 3x_2 + 5x_3 \geq 2$ $3x_1 + x_2 + 7x_3 \leq 3$ $x_1 + 4x_2 + 6x_3 \leq 5$ x_1, x_2, x_3 all ≥ 0	16	3	3

		UNIT - III			
5	a	Determine how much quantity should be stepped from factory to various destinations for the following transportation problem using the least cost method: Destination / Capacity Factory: *T* 6 8 8 5 — 30; *B* 5 11 9 7 — 40; *M* 8 9 7 13 — 50 Demand: 35 28 32 25	06	2	2
	b	The following table provides all the necessary information on the availability of supply to each warehouse, the requirement of each market, and the unit transportation cost (in Rs.) from each warehouse to each market. Market / Supply Warehouse: *A* 6 3 5 4 — 22; *B* 5 9 2 7 — 15; *C* 5 7 8 6 — 8 Demand: 7 12 17 9 The shipping clerk of the shipping agency has worked out the following schedule, based on his own experience: 12 units from A to Q, 1 unit from A to R, 9 units from A to S, 15 units from B to R, 7	10	3	3

units from C to P and 1 unit from C to R.

a. Check and see if the clerk has the optimal schedule.
b. Find the optimal schedule and minimum total transport cost.

OR

6 a. There are four machines and four operators with their respective payoffs (in 100 rupees) as shown below. How the machines should be allocated to each operator so as to minimize the total pay – off. (08, 3, 3)

Machines	Operators O_1	O_2	O_3	O_4
M_1	5	4	2	7
M_2	7	8	3	5
M_3	5	3	4	6
M_4	4	6	6	5

b. Five salesmen are to be assigned to five districts. Estimates of sales revenue (in thousands) for each salesman are given as follows: (08, 3, 3)

	A	B	C	D	E
1	32	38	40	28	40
2	40	24	28	21	36
3	41	27	33	30	37
4	22	38	41	36	36
5	29	33	40	35	39

Find the assignment pattern that maximizes the sales revenue.

UNIT - IV

7 a. Compare CPM and PERT with respect to: (04, 2, 3)

i. Orientation
ii. Model
iii. Estimates
iv. Crashing concept

b. Draw the network diagram and determine the early start, finish and late start, finish in respect of all node points and identify critical path for the following: (12, 3, 3)

Activity	1 – 2	1 – 3	1 – 4	2 – 5	3 – 7	4 – 6	5 – 7
Duration	10	8	9	8	16	7	7
Activity	5 – 8	6 – 7	6 – 9	7 – 10	8 – 10	9 – 10	
Duration	6	7	5	12	13	15	

OR

8 a. Explain the three estimates of Program Evaluation & Review Technique. (04, 3, 3)

b. The following network diagram represents activities associated with a project: (12, 4, 4)

Activity	A	B	C	D	E	F	G	H	I
Optimistic Time	5	18	26	16	15	6	7	7	3
Pessimistic Time	10	22	40	20	25	12	12	9	5
Most Likely Time	8	20	33	18	20	9	10	8	4

i. Find expected completion time and variance of each activity
ii. Draw Network diagram
iii. The critical path and project duration
iv. The duration of the project that will have 95 per cent chance of being completed.

		UNIT - V			
9	a	Use the principle of dominance to find the optimal strategy for both the players and also value of the game for the following pay – off matrix: Player B I II III IV Player A: 1 (4 2 3 6); 2 (3 4 7 5); 3 (6 3 5 4)	08	4	4
	b	Use the graphical method to find the optimal strategy for both the players and also value of the game for the following pay – off matrix: Player B B1 B2 B3 Player A: A1 (1 3 12); A2 (8 6 2)	08	2	4
		OR			
10	a	List the properties of a game theory.	06	2	2
	b	Solve the following game for optimal strategy for both the players and value of the game by dominance property whose payoff matrix is: B I II III IV V VI A I (4 2 0 2 1 1); II (4 3 1 3 2 2); III (4 3 7 -5 1 2); IV (4 3 4 -1 2 2); V (4 3 3 -2 2 2)	10	3	3